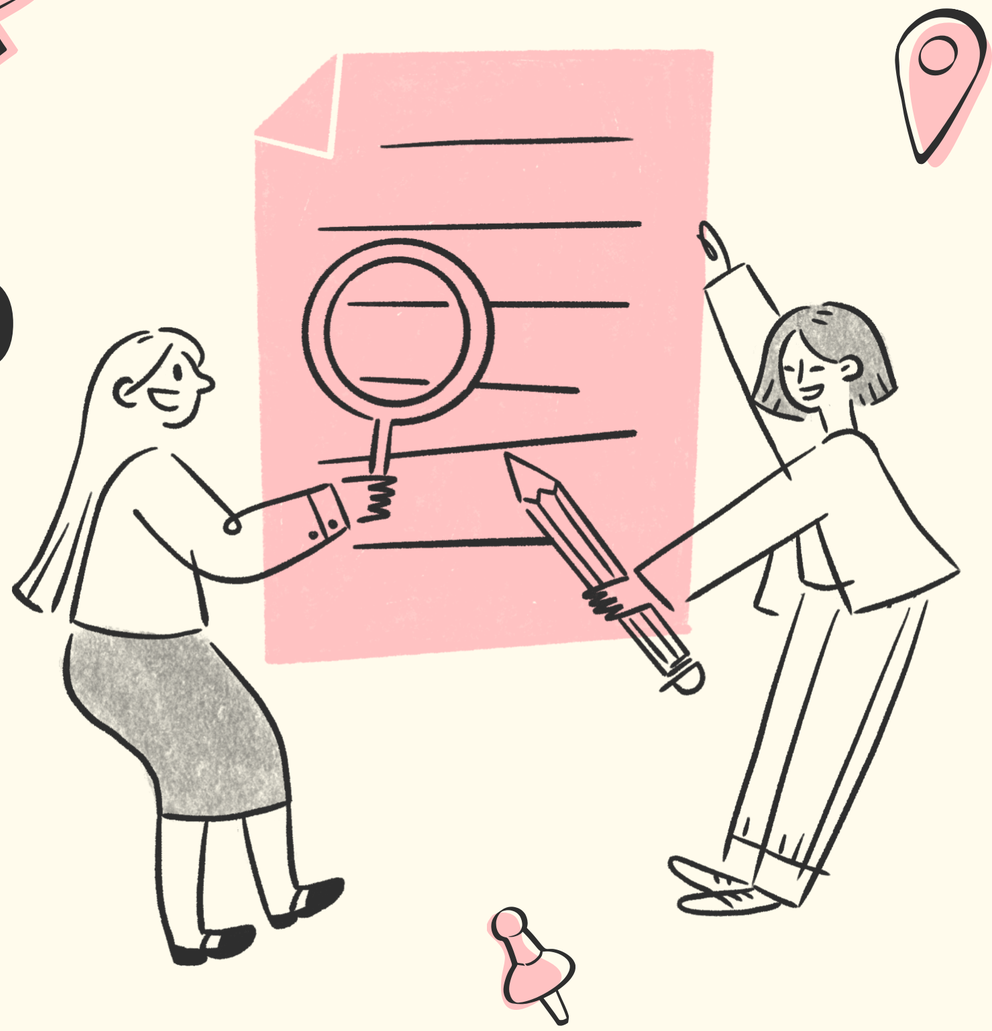
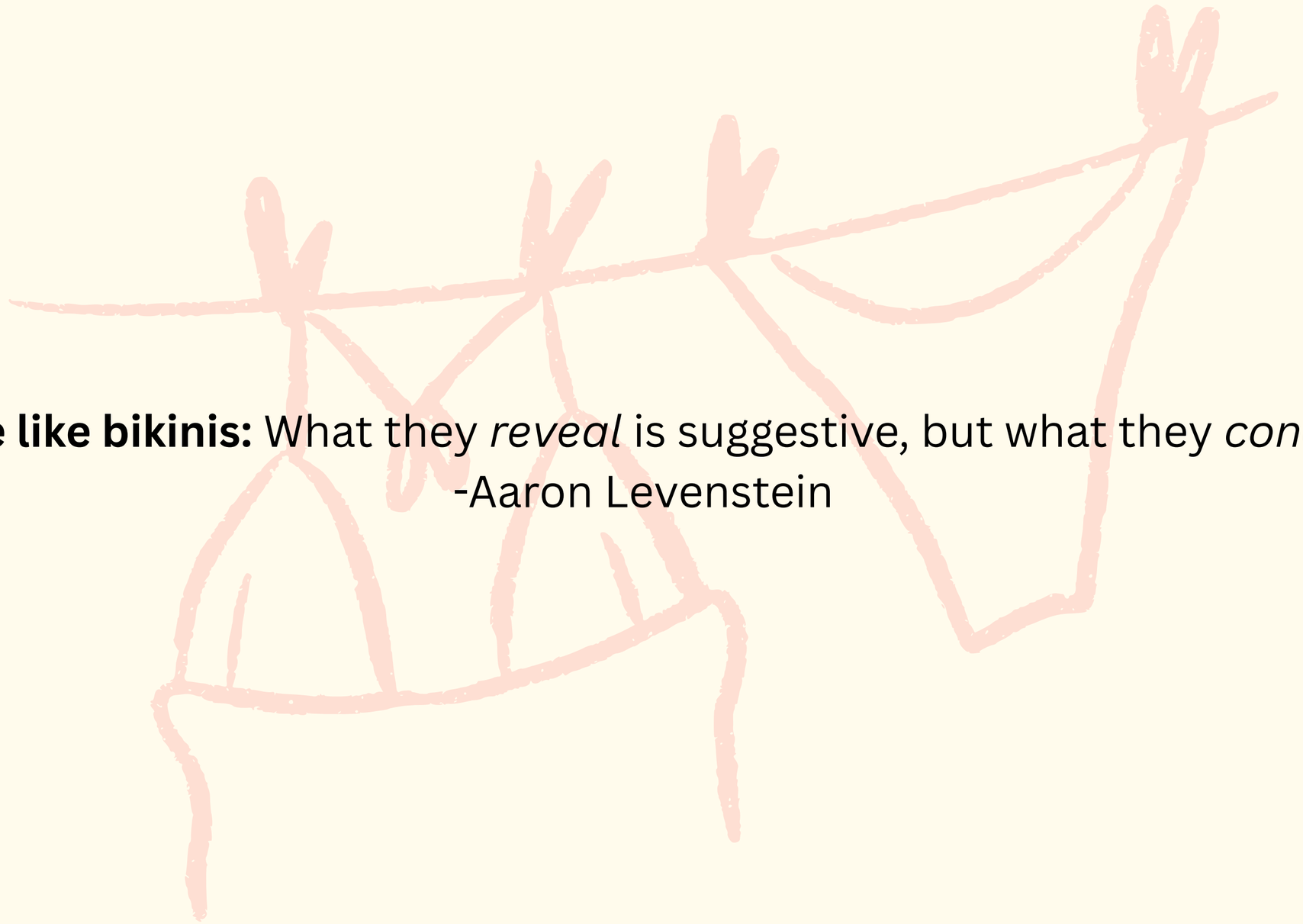
Dalmatian Group Presentation #1

Yu Xin, Shi Han, Janelle, Audris





“**Statistics are like bikinis:** What they *reveal* is suggestive, but what they *conceal* is vital.”
-Aaron Levenstein

Overview



Introduction
Objectives
Methodology
Result and Analysis
Conclusion

Introduction

The **How Couples Meet and Stay Together (HCMST)** 2017–2020–2022 dataset studied how Americans meet their spouses and romantic partners

We selected this dataset to investigate the relationship between **socioeconomic status (SES)**, **racial composition of couples**, and **relationship stability.**

Main objectives



(I) Operationalise socioeconomic status (SES)

- Use household income and educational attainment to capture respondents' socioeconomic position

(2) Measure relationship stability longitudinally

- Examine whether relationships persisted or dissolved across waves, alongside relationship duration

(3) Examine race as a moderating variable

- Assess whether the association between SES and relationship stability differs across racial groups

Our research question

To what extent does the association between **socioeconomic status** and **relationship stability** vary across **racial groups** in the US?

Hypothesis

To what extent does the association between **socioeconomic status** and **relationship stability** vary across **racial groups** in the US?

Methodology

Our research question

Independent variable

Dependent variable

To what extent does the association between **socioeconomic status** and **relationship stability** vary across **racial groups** in the US?

Moderator

Methodology (SES)

Indicators used

- Household income (ppincimp)
- Respondent education (ppeduc)
- Partner education (q10)

What we did

- Converted 19 household-income brackets into an ordinal scale **(1–19)**
- Grouped income into **Low / Middle / High** tertiles
- Collapsed respondent and partner education into **Less than HS / HS / Some college / Bachelor's+**
- Created an education comparison variable: **Respondent higher / Partner higher / Same**
- Treated missing/refused partner-education responses as **missing**

This was to transform the original survey responses into consistent measures of SES for examining its association with relationship stability.

Methodology (Relationship Stability)

Indicators used

- Wave 2 and Wave 3 relationship information
- Relationship duration (how_long_relationship)

What we did

- Combined later-wave information into **relationship_stable**
  - **1 = stable**
  - **0 = dissolved**
- Created **relationship_duration_years** as the standardised relationship duration
- Kept unclassifiable cases as missing
- Checked missingness and invalid/outlier values

This was to convert longitudinal survey responses into consistent measures of relationship stability, allowing us to compare whether relationships persisted or dissolved over time and account for differences in relationship duration.

Methodology (Race)

Indicators used

- Respondent race/ethnicity across Waves I–3:
  - ppethm
  - pp2_ppethm
  - pp3_ppethm
- Partner race (q6b)

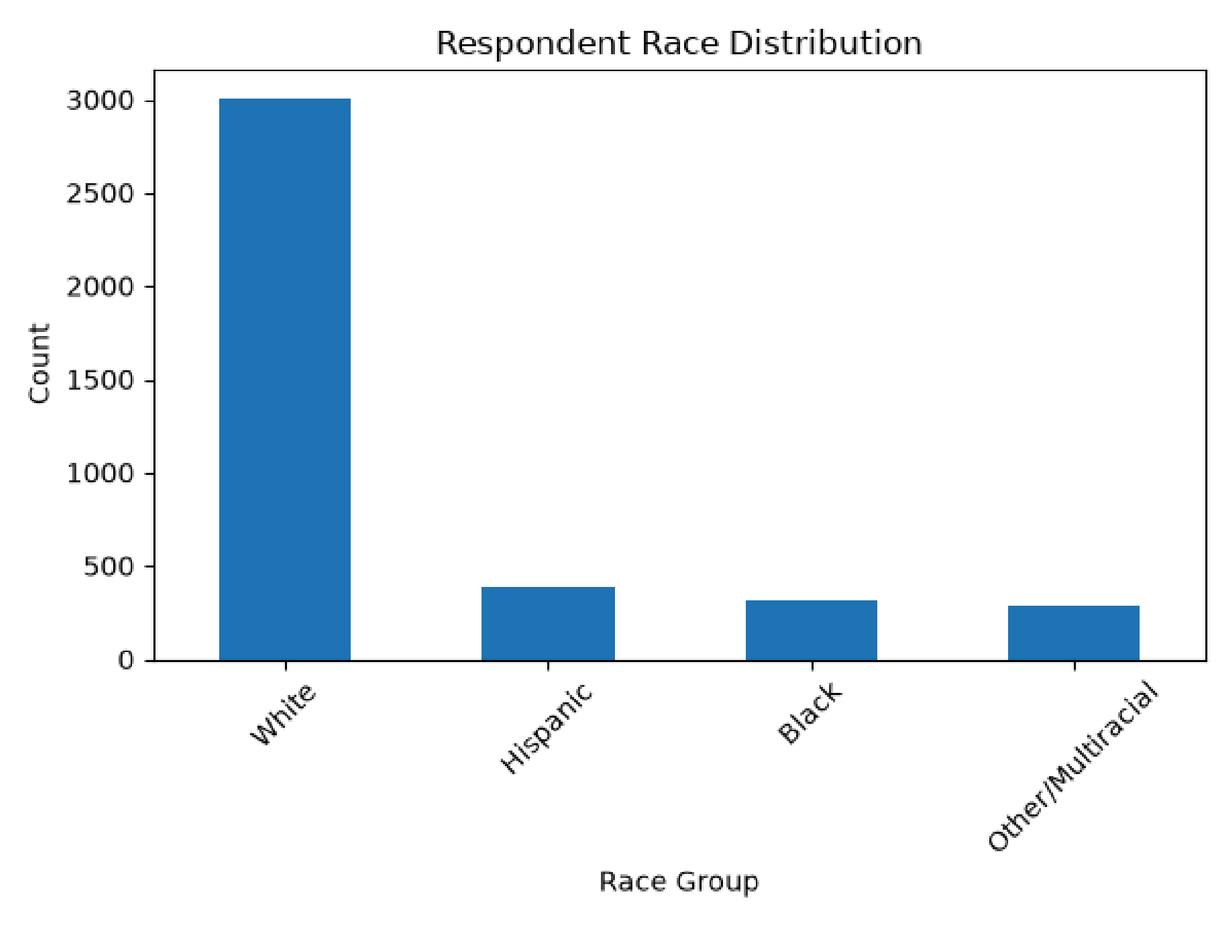
What we did

- Standardised respondent race labels
- Created one respondent race variable using the earliest available response (i.e wave I → wave 2 if missing → wave 3 if still missing)
- Collapsed respondent race into **White, Black, Hispanic, Other/Multiracial**
- Recoded partner race into **White, Black, Other/Multiracial**
- Treated refused partner-race responses as **missing**

This was to create consistent racial categories for examining whether the association between SES and relationship stability differs across racial groups.

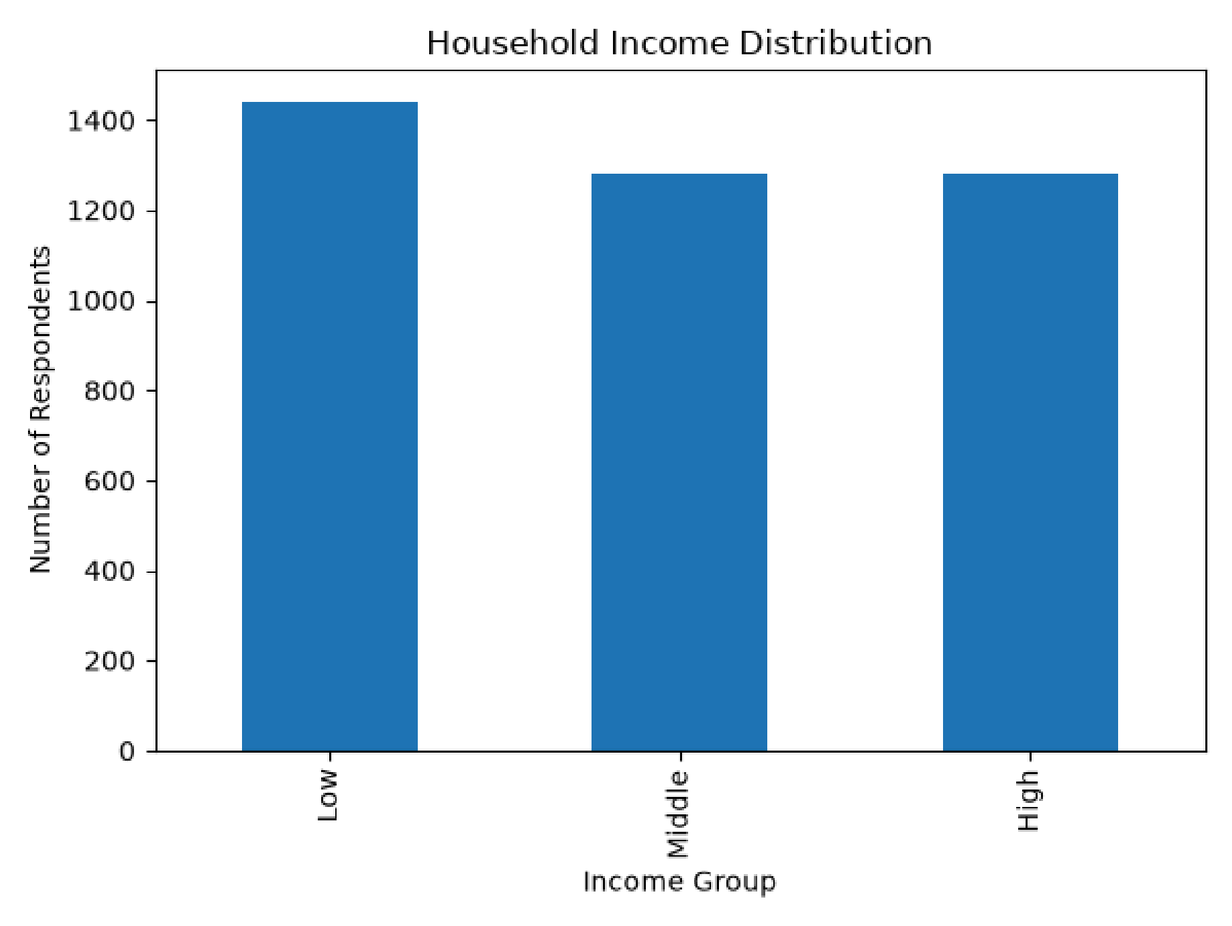
Result & Analysis

Sample Overview (Race)



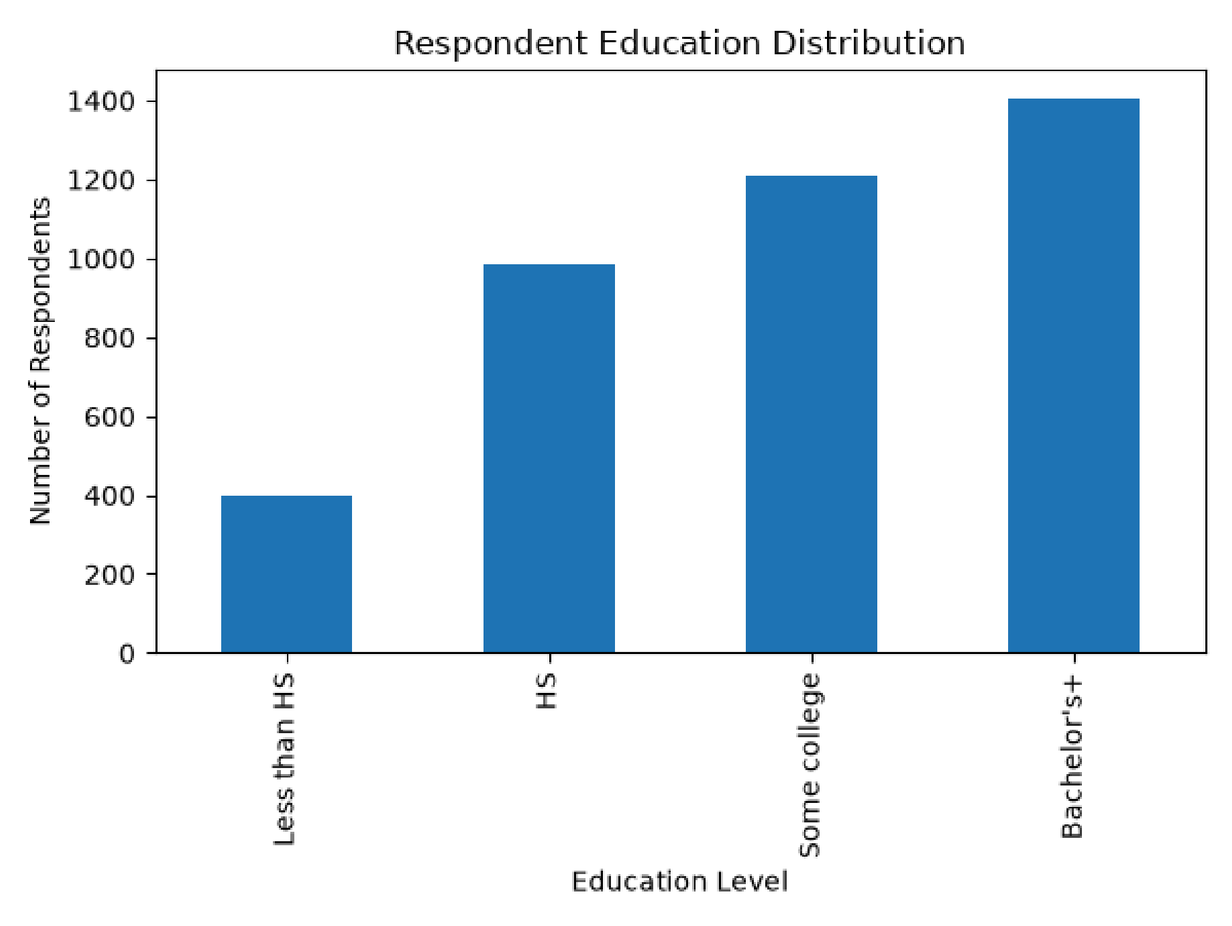
Sample is heavily White (~3,000), with smaller Hispanic (~390), Black (~320), and Other/Multiracial (~290) groups

Sample Overview (Income)



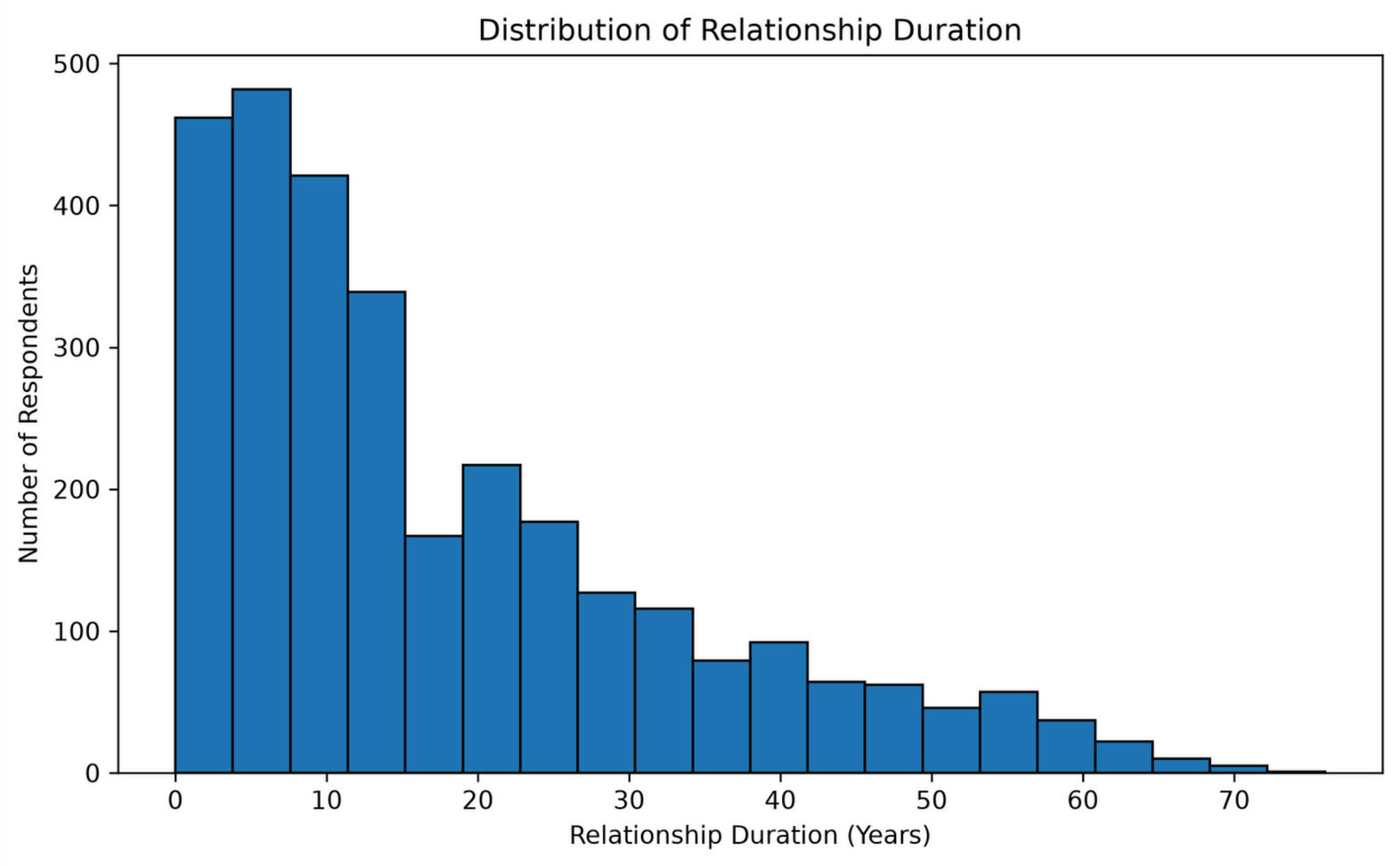
Sample is fairly balanced across Low (~1,440), Middle (~1,280), High (~1,280)

Sample Overview (Education)



Skews toward higher education: most respondents have “Some college” or “Bachelor’s+” (Partner Education is similarly distributed)

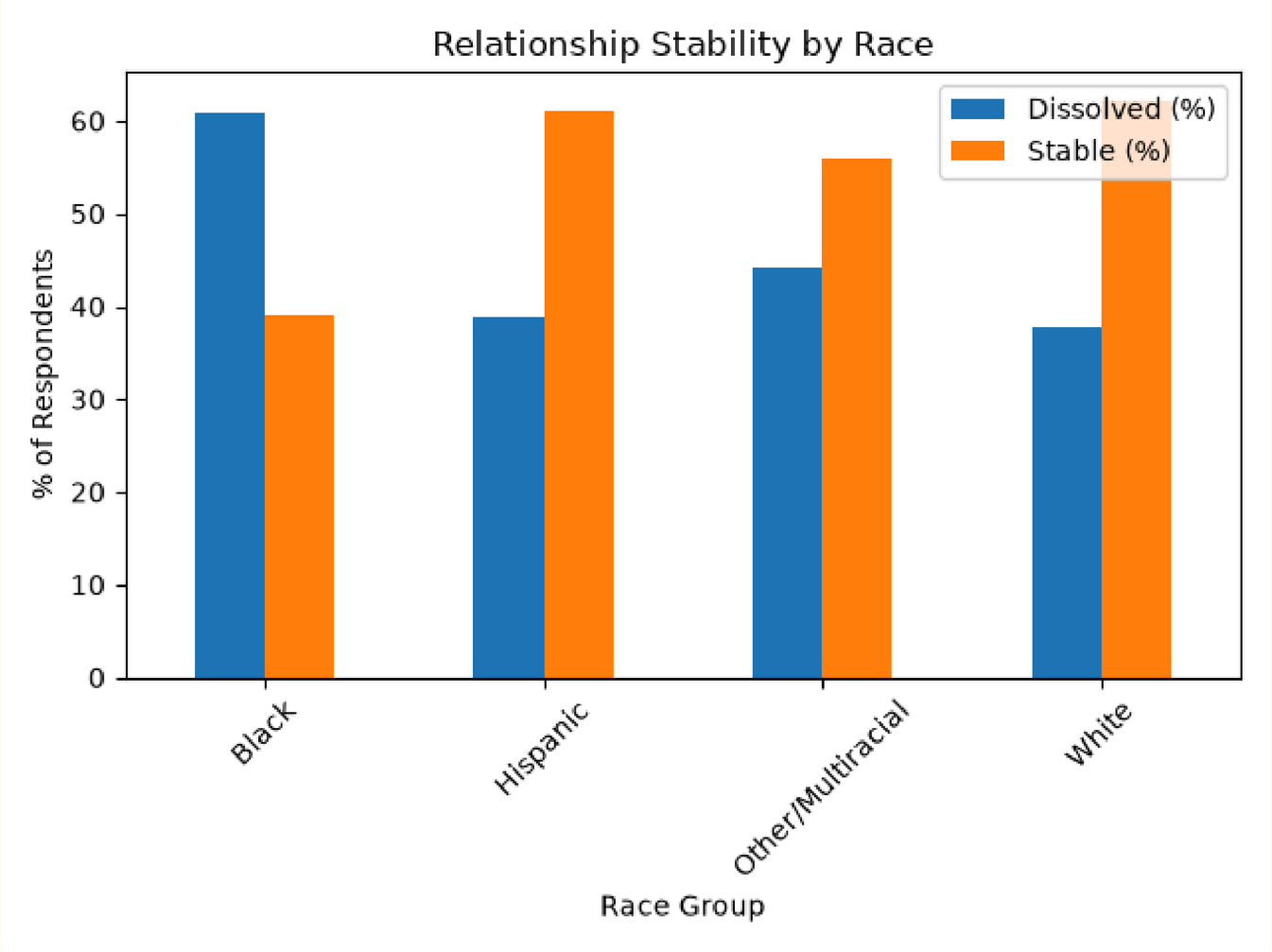
Sample Overview (Relationship duration)



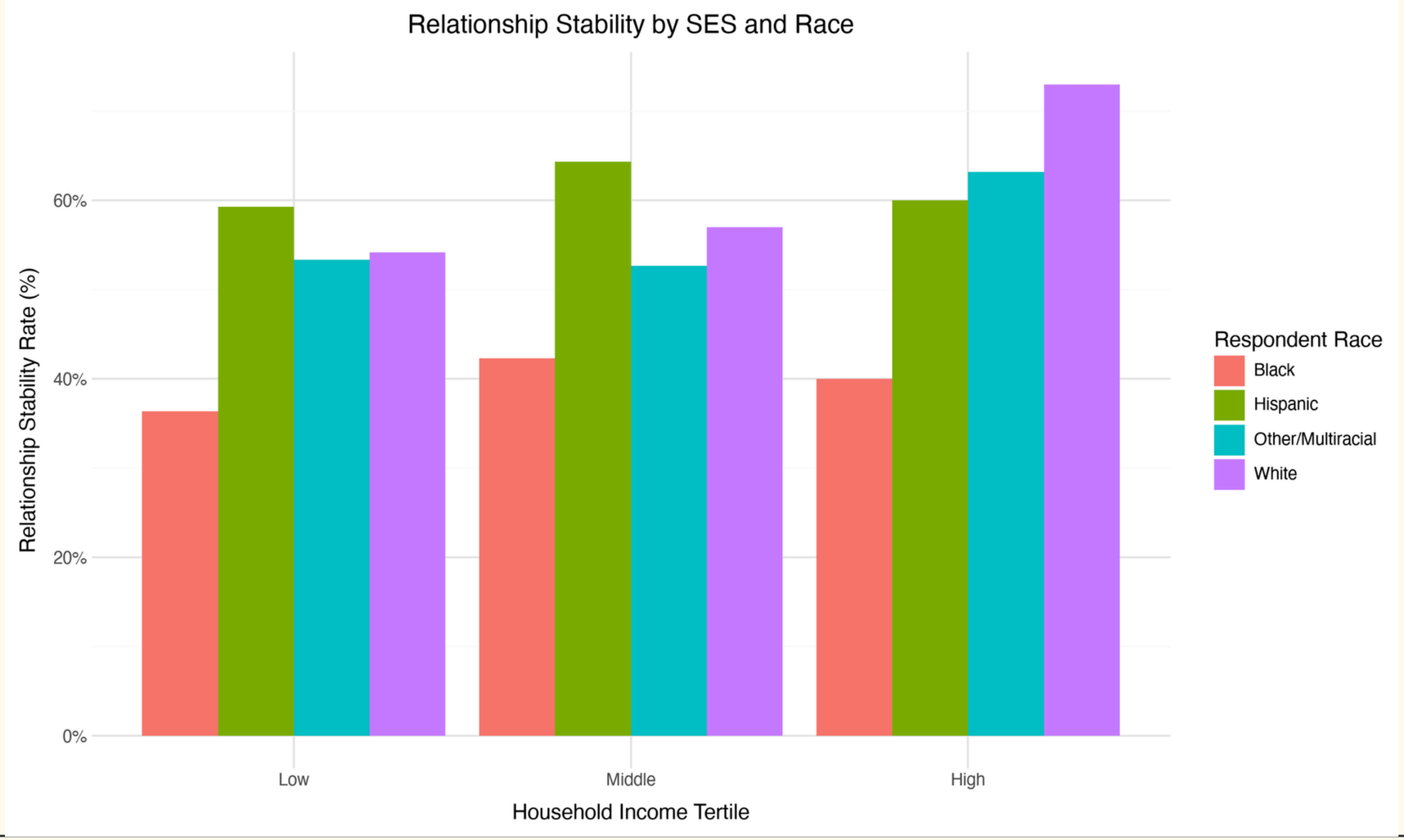
Right-skewed — most relationships are under 15 years, with a long tail out to 60-70+ years

Race x Stability interaction

Black respondents show significantly lower stability than all other groups, a ~20+ point gap



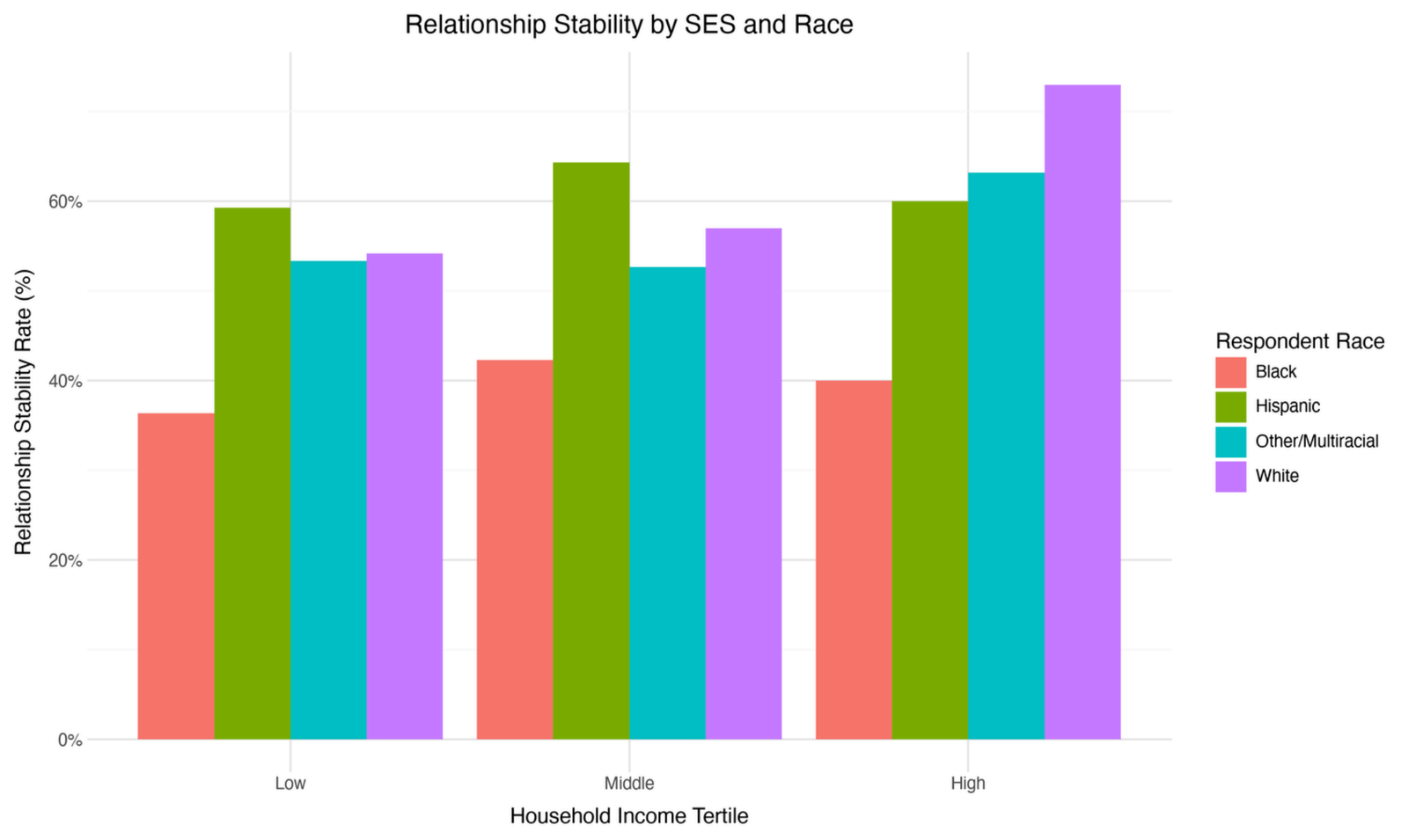
SES × Race Interaction



Patterns Observed

- Blacks:
  - Consistently **lower** relationship stability rates across *all three income levels*

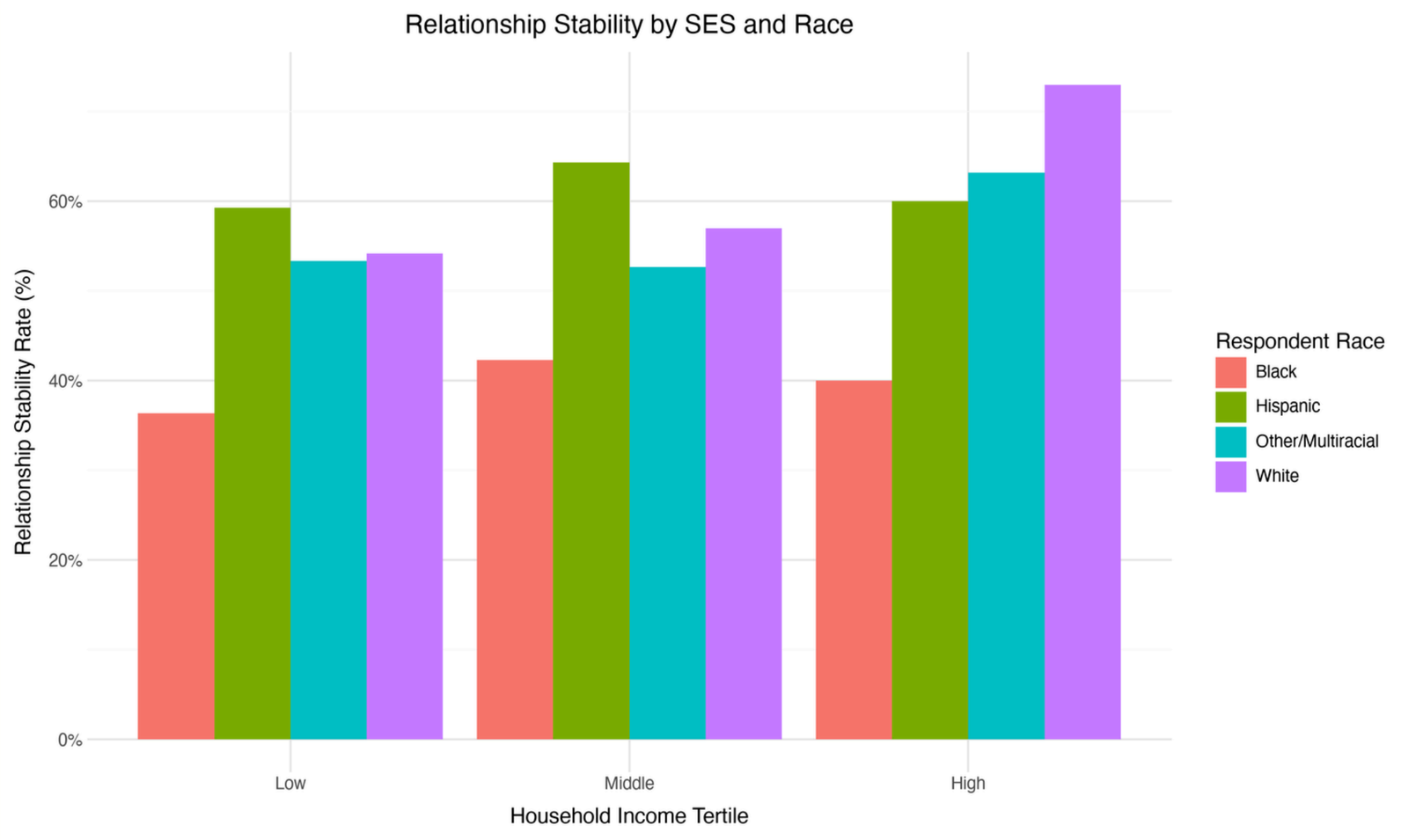
- Whites:
  - **Strong positive relationship** between income and relationship stability



Patterns Observed

- Hispanics:
  - Relative **high relationship stability across all SES levels**

- Others:
  - **Moderate positive relationship** between income and relationship stability



Overview:



What's done?

- Data recoding
- Data cleaning
- Descriptive analytics
- Brief look at interaction





Overview:



What's next?

Logistic Regression Model

- Check for statistical significance
- Adding an interaction term to calculate exact p-values to evaluate RQ
- Suitable because of binary DV (relationship stablity)



Linking findings to theory

- Social Exchange Theory
- Comparison Level for Alternatives (CLalt)



Looking at secondary SES measures

Limitations:



Limitations:

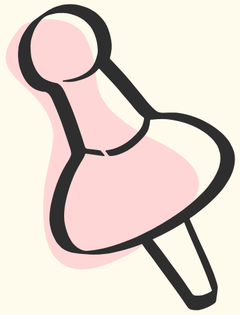
- Attrition bias
- Partner race mismatch

Actionables:

- Refining racial categories
- Inspect how many respondents were lost to attrition and adjust











Thank You



